

Bryan M. Maitland, Ph.D.

Research Fish Biologist

Curriculum Vitae

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Rocky Mountain Research Station,
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Appointments

2024–Present **Research Fish Biologist**, Rocky Mountain Research Station, USDA Forest Service
2023–2024 **Research Associate**, Dept. of Civil and Environmental Engineering, University of Wisconsin-Madison
2022–2023 **Postdoctoral Fellow**, Center for Limnology, University of Wisconsin-Madison
2020–2022 **WI Water Resources Postdoctoral Fellow**, Aquatic Sciences Center (UW-Madison) and WDNR

Academic Affiliations

2026–Present **Affiliate Faculty**, Dept. of Biology, University of Tennessee at Chattanooga

Education

2020 Ph.D. Ecology University of Wyoming
2015 M.Sc. Conservation Biology (Fisheries) University of Alberta
2011 B.S. Conservation Biology *cum laude* SUNY: College of Environmental Science and Forestry

Research summary

I am a conservation ecologist studying how disturbance—wildfire, climate change, and altered flow regimes—propagates through stream ecosystems and the food webs they support. I combine stable isotope ecology, otolith and tissue chemistry, long-term agency biomonitoring records, remote sensing, and spatial and Bayesian modeling; much of my published work integrates multi-decade state and federal monitoring datasets with spatially explicit habitat and climate data to separate direct from indirect drivers of community change. I am a partner on the JFSP-funded project *From flames to fish: co-management for wildfire and aquatic species in Idaho* (2026–2028), I co-chaired the AGU session *Science to Restore and Manage Wildland Aquatic Habitats and Watersheds* (2024), and the *Fish and Fire* symposium at the 2026 AFS Annual Meeting. A through-line in my work is building open, reusable tools and datasets—R packages, databases, and reproducible workflows. I also mentor graduate and undergraduate researchers, including Garrison Ferone at the University of Idaho.

- Since 2014 I have authored 26 peer-reviewed journal articles and 1 R package (3822 downloads), and presented 47 papers and posters at AFS, SFS, ASLO, AGU, IAGLR, and regional meetings.
- Google Scholar h-index 14, 739 total citations (as of 7 October 2026).
- Since 2015 I have led, co-led, or partnered on \$0.3 million in funded projects, and contributed to invited proposals exceeding \$2 million in scope.
- I have reviewed 60 manuscripts for 19 journals since 2016, and served as a grant reviewer for NSF (Division of Environmental Biology) and the Lake Champlain Fisheries Research Program.

Selected publications

Underlined names indicate undergraduate research mentees. Citation counts from Google Scholar (7 October 2026).

8. Sparks*, M.M., Maitland*, B.M., Felts, E., Swartz, A., & Frater, P. (2026). hatchR: A toolset to predict when fish hatch and emerge. *Fisheries*, 51(2), 49–59. <https://doi.org/10.1093/fshmag/vuaf078> [Citations: 3].
7. Maitland, B.M., Bootsma, H.A., Bronte, C.R., Bunnell, D.B., Feiner, Z.S., Fenske, K.H., Fetzner, W.W., Foley, C.J., Gerig, B.S., Happel, A., Höök, T.O., Keppeler, F.W., Kornis, M.S., Lepak, R.F., McNaught, A.S., ... Jensen, O.P. (2024). Testing food web theory in a large lake: The role of body size in habitat coupling in Lake Michigan. *Ecology*, 105(10), e4413. <https://doi.org/10.1002/ecy.4413> [Citations: 15].
6. Maitland, B.M., & Latzka, A.W. (2022). Shifting climate conditions affect recruitment in Midwestern stream trout, but depend on seasonal and spatial context. *Ecosphere*, 13(12), e4308. <https://doi.org/10.1002/ecs2.4308> [Citations: 17].
5. Kirk, M.A., Maitland, B.M., Hickerson, B.T., Walters, A.W., & Rahel, F.J. (2022). Climatic drivers and ecological impacts of a rapid range expansion by non-native smallmouth bass. *Biological Invasions*, 24, 1311–1326. <https://doi.org/10.1007/s10530-021-02724-z> [Citations: 30].
4. Maitland, B.M., & Rahel, F.J. (2023). Aquatic food web expansion and trophic redundancy along the Rocky Mountain–Great Plains ecotone. *Ecology*, 104(7), e4103. <https://doi.org/10.1002/ecy.4103> [Citations: 14].
3. Barrus, N.T., Maitland, B.M., & Rahel, F.J. (2024). Assessing a standardized method to identify optimal baselines for trophic position estimation in stable isotope studies of stream ecosystems. *Hydrobiologia*, 851, 4673–4691. <https://doi.org/10.1007/s10750-024-05618-y> [Citations: 3].
2. Maitland, B.M., Poesch, M., Anderson, A.E., & Pandit, S.N. (2016). Industrial road crossings drive changes in community structure and instream habitat for freshwater fishes in the boreal forest. *Freshwater Biology*, 61, 1–18. [Citations: 44].

1. Craig, L.S., Olden, J.D., Arthington, A.H., Entekhabi, S., Hawkins, C.P., Kelly, J.J., Kennedy, T.A., **Maitland, B.M.**, Rosi, E.J., Roy, A.H., Strayer, D.L., Tank, J.L., West, A.O., & Wooten, M.S. (2017). Meeting the challenge of interacting threats in freshwater ecosystems: A call to scientists and managers. *Elementa: Science of the Anthropocene*, 5(72), 1–15. <https://doi.org/10.1525/elementa.256> [Citations: 210].

Selected funding

2026–2028	Partner. “FLAMES: From flames to fish: Co-management for wildfire and aquatic species in Idaho”. <i>Joint Fire Science Program</i> .	\$150,284
2025–2027	Lead-PI. “Monitoring Chinook Salmon spawning from space”. <i>Trout Unlimited</i> .	\$38,122
2024–2027	Lead-PI. “Chinook Salmon life history and otolith microchemistry”. <i>UC-Davis and USDA Forest Service</i> .	\$50,184
Pending	Co-PI. “Idaho Bull Trout migration timing and habitat use”. <i>U.S. Fish and Wildlife Service</i> .	\$159,900
Pending	Lead-PI. “From reservoir to redd: linking reservoir operations to bull trout demography”. <i>U.S. Bureau of Reclamation</i> .	\$145,000
Pending	Co-PI. “FLAMES: Development of fire-stream exposure metrics to improve the assessment of aquatic effects using geospatial data”. <i>Joint Fire Science Program</i> .	\$39,952
Pending	Lead-PI. “Idaho Bull Trout food web ecology and hybrid interactions”. <i>U.S. Fish and Wildlife Service</i> .	\$20,000

Selected awards and honors

2026	Outstanding Service Award, USDA Forest Service, Rocky Mountain Research Station
2024	Outstanding Service Award, USDA Forest Service, Rocky Mountain Research Station
2021	Best Professional Presentation Award, Wisconsin Chapter American Fisheries Society
2020	Wisconsin Water Resources Postdoctoral Fellow, University of Wisconsin-Madison
2019	Dean's Graduate Fellowship, College of Arts and Sciences, University of Wyoming
2018	Eugene Maughan Graduate Award, Western Division American Fisheries Society

Mentoring and teaching

2023–present	Supervised 3 graduate/post-baccalaureate researchers (U. Idaho, UW-Madison, UT-Chattanooga); MS committee member, UT-Chattanooga
2015–2025	Mentored 11 undergraduate researchers, two of whom have led or co-authored peer-reviewed publications
2014–2020	Instructor of record for Ichthyology (UWyo); teaching assistant for 12 sections of Ichthyology and Fisheries Management (UWyo, U. Alberta)

Selected service and synergistic activities

2025–2029	Mentor of Practice, NSF IRES: Climate Change Resilience in Aquatic Bright Spots
2026	Program co-chair, <i>Fish and Fire</i> symposium, AFS Annual Meeting
2024	Program co-chair, <i>Science to Restore and Manage Wildland Aquatic Habitats and Watersheds</i> , AGU Fall Meeting
2020–2022	Member, Trout Species Team, Wisconsin Dept. of Natural Resources; Fisheries Working Group, Wisconsin Initiative on Climate Change Impacts
2015–present	Member, Freshwater Working Group, Society for Conservation Biology; AFS, SCB, ESA, SFS

Developer and maintainer of *hatchR*, an open-source R package for predicting fish hatch and emergence timing from thermal regimes. Instructor for R programming and data visualization workshops at American Fisheries Society chapter meetings. Research covered by regional press and public radio.